

Unit 3: Knowledge Representation and Probabilistic Reasoning

SubUnit 3.1: Knowledge-based Agents

Kiran Bagale

IOE, Thapathali Campus

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Outline

- 1 Introduction to Knowledge-based Agents
- 2 Components of Knowledge-based Agents
- 3 The Wumpus World Example
- 4 Building a Knowledge-based Agent
- 5 Types of Knowledge
- 6 Knowledge Representation Schemes
- 7 Advantages and Challenges
- 8 Comparison with Other Agent Types
- 9 Conclusion and Future Directions

What is a Knowledge-based Agent?

Definition

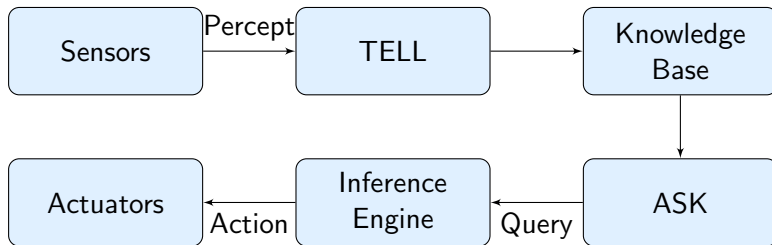
A knowledge-based agent is an agent that reasons by operating on internal representations of knowledge.

Key Characteristics:

- Contains a **knowledge base** (KB) - a set of sentences
- Uses a **knowledge representation language**
- Performs **inference** to derive new sentences
- Makes decisions based on knowledge about the world

Reference: Russell & Norvig (2020), Chapter 7

Architecture of Knowledge-based Agents



Two fundamental operations:

- **TELL:** Add new sentences to the knowledge base
- **ASK:** Query the knowledge base for what is known

The Knowledge Base

What is a Knowledge Base?

- A set of **sentences** expressed in a knowledge representation language
- Each sentence is called an **axiom** when it's given without derivation
- Represents facts about the world

Operations on KB:

- `TELL(KB, sentence)`: Adds new sentence to KB
- `ASK(KB, query)`: Returns an answer from KB
- `RETRACT(KB, sentence)`: Removes sentence from KB

Example

`TELL(KB, "It is raining")`

`ASK(KB, "Is it raining?") → True`

Requirements for a KR Language:

- 1 **Syntax:** Defines how to construct valid sentences
- 2 **Semantics:** Defines the meaning of sentences
- 3 **Inference rules:** Derives new sentences from existing ones

Desirable Properties:

- **Expressiveness:** Can represent complex knowledge
- **Conciseness:** Compact representation
- **Efficiency:** Fast inference
- **Naturalness:** Close to human reasoning

Reference: Rich, Knight & Nair (2009), Chapter 5

What is Inference?

- Process of deriving new sentences from existing ones
- Uses **logical consequence** and **entailment**

Entailment

$KB \models \alpha$ means:

Sentence α is **entailed** by KB if α is true in all worlds where KB is true

Inference Properties:

- **Soundness:** Derives only entailed sentences
- **Completeness:** Can derive any entailed sentence

Example

If KB contains: "All humans are mortal" and "Socrates is human"

Then we can infer: "Socrates is mortal"

Inference Aspect

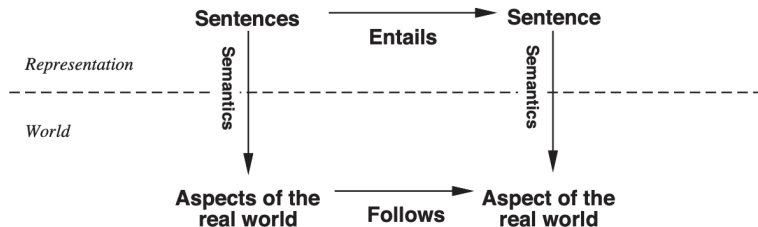


Figure: Sentences are physical configurations of the agent, and reasoning is a process of constructing new physical configurations from old ones.

The final issue to consider is grounding—the connection between logical reasoning processes and the real environment in which the agent exists. In particular, how do we know that KB is true in the real world? (After all, KB is just “syntax” inside the agent’s head.)

Logical reasoning should ensure that the new configurations represent aspects of the world that actually follow from the aspects that the old configurations represent.

The Wumpus World: Problem Description

Environment:

- 4×4 grid of rooms
- Agent starts at square [1,1]
- Goal: Find gold and return to start

Hazards:

- **Wumpus:** Monster that kills the agent (one Wumpus)
- **Pits:** Fall into them and die (multiple pits)

Percepts:

- **Stench:** In squares adjacent to Wumpus
- **Breeze:** In squares adjacent to pit
- **Glitter:** In square with gold
- **Bump:** When hitting wall
- **Scream:** When Wumpus is killed

Reference: Russell & Norvig (2020), Section 7.2

A typical Wumpus World: Example

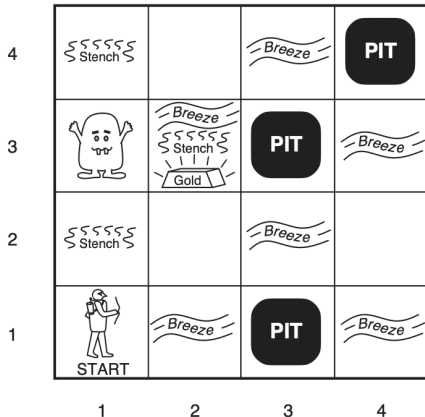


Figure: A typical wumpus world. The agent is in the bottom left corner, facing right.

The Wumpus World: Agent Actions

Available Actions:

- Forward: Move forward one square
- TurnLeft: Turn left 90°
- TurnRight: Turn right 90°
- Grab: Pick up gold if in same square
- Shoot: Fire arrow forward (only one arrow)
- Climb: Leave the cave (only from [1,1])

Performance Measure:

- +1000 for finding gold
- -1000 for falling into pit or being eaten
- -1 for each action taken
- -10 for using the arrow

Wumpus World: Knowledge Representation

Representing the World State:

- $P_{x,y}$: There is a pit in $[x,y]$
- $W_{x,y}$: The Wumpus is in $[x,y]$
- $B_{x,y}$: Agent perceives breeze in $[x,y]$
- $S_{x,y}$: Agent perceives stench in $[x,y]$

Example Rules:

- $B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})$
- If breeze at $[1,1]$, then pit at $[1,2]$ or $[2,1]$
- $S_{1,1} \Rightarrow (W_{1,2} \vee W_{2,1})$
- If stench at $[1,1]$, then Wumpus at $[1,2]$ or $[2,1]$

Safe Squares:

- $\neg B_{1,1} \Rightarrow \neg P_{1,2} \wedge \neg P_{2,1}$
- No breeze means adjacent squares are safe

A typical Wumpus World: Example

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2	2,2	3,2	4,2
OK			
1,1	2,1	3,1	4,1
A			
OK	OK		

- A

 = Agent
- B = Breeze
- G = Glitter, Gold
- OK = Safe square
- P = Pit
- S = Stench
- V = Visited
- W = Wumpus

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2	2,2	3,2	4,2
OK	P?		
1,1	2,1	3,1	4,1
V	A	P?	
OK	B		
	OK		

(a)

(b)

Figure: The first step taken by the agent in the wumpus world. (a) The initial situation, after percept [None, None, None, None, None]. (b) After one move, with percept [None, Breeze, None, None, None].

Wumpus World: Reasoning Example

Scenario:

- 1 Agent at [1,1], perceives no stench, no breeze
- 2 Agent moves to [2,1], perceives breeze
- 3 What can the agent infer?

Inference Process:

- From [1,1]: $\neg B_{1,1}$ implies [1,2] and [2,1] are safe (no pits)
- From [2,1]: $B_{2,1}$ implies pit in [2,2] or [3,1] or [1,1]
- Since [1,1] is known safe, pit must be in [2,2] or [3,1]
- Agent knows [1,2] is safe to visit
- Agent uncertain about [2,2] and [3,1]

Key Insight

Agent must reason under **partial observability** and **uncertainty**

A typical Wumpus World: Example

1,4	2,4	3,4	4,4
1,3 W!	2,3	3,3	4,3
1,2 A S OK	2,2 OK	3,2	4,2
1,1 V OK	2,1 B V OK	3,1 P!	4,1

(a)

A = Agent
B = Breeze
G = Glitter, Gold
OK = Safe square
P = Pit
S = Stench
V = Visited
W = Wumpus

1,4	2,4 P?	3,4	4,4
1,3 W!	2,3 A S G B	3,3 P?	4,3
1,2 S V OK	2,2 V OK	3,2	4,2
1,1 V OK	2,1 B V OK	3,1 P!	4,1

(b)

Figure: Two later stages in the progress of the agent. (a) After the third move, with percept [Stench, None, None, None, None]. (b) After the fifth move, with percept [Stench , Breeze , Glitter , None , None].

Generic Knowledge-based Agent Algorithm

KB-Agent Pseudocode

```
function KB-AGENT(percept) returns an action
  persistent: KB (knowledge base)
               t (time counter, initially 0)

  TELL(KB, MAKE-PERCEPT-SENTENCE(percept, t))
  action <- ASK(KB, MAKE-ACTION-QUERY(t))
  TELL(KB, MAKE-ACTION-SENTENCE(action, t))
  t <- t + 1
  return action
```

Key Steps:

- 1 Tell KB about percept
- 2 Ask KB what action to perform
- 3 Tell KB about chosen action
- 4 Update time and return action

Levels of Knowledge Representation

1. Epistemological Level:

- What facts are represented
- Example: "The Wumpus is in room [2,2]"

2. Logical Level:

- Encoding of facts in formal language
- Example: $W_{2,2}$ (propositional logic)

3. Implementation Level:

- Data structures and algorithms
- Example: Boolean array, inference engine

Reference: Rich, Knight & Nair (2009), Section 5.1

A typical Wumpus World: logic

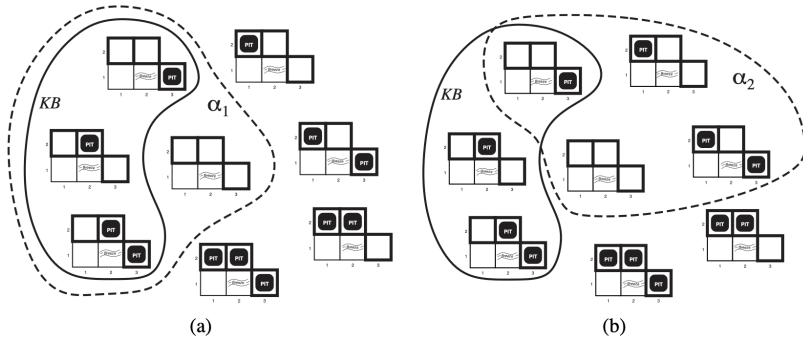


Figure: Possible models for the presence of pits in squares [1,2], [2,2], and [3,1]. The KB corresponding to the observations of nothing in [1,1] and a breeze in [2,1] is shown by the solid line. (a) Dotted line shows models of 1 (no pit in [1,2]). (b) Dotted line shows models of 2 (no pit in [2,2]).

Designing the Knowledge Base

What to Include in KB:

① Domain Knowledge:

- Rules about how the world works
- Physics, causality, typical behaviors

② Specific Facts:

- Current state information
- Observations and percepts

③ Goal Information:

- Desired states
- Performance criteria

④ Action Models:

- Preconditions and effects
- Action costs

Categories of Knowledge

1. Declarative Knowledge (Know-what):

- Facts and relationships
- Example: "Water boils at 100°C"
- Represented as logical sentences

2. Procedural Knowledge (Know-how):

- How to do things
- Example: "To make tea, boil water, add tea leaves..."
- Represented as procedures or rules

3. Meta-knowledge (Know-about-knowledge):

- Knowledge about what you know
- Example: "I know that I don't know quantum physics"
- Important for reasoning about uncertainty

Types of Reasoning

1. Deductive Reasoning:

- Derive logically certain conclusions
- Example: All birds have wings; Tweety is a bird $\Rightarrow$ Tweety has wings
- **Sound and complete**

2. Inductive Reasoning:

- Generalize from specific observations
- Example: Observe 100 white swans $\Rightarrow$ All swans are white
- **Not guaranteed to be correct**

3. Abductive Reasoning:

- Infer best explanation for observations
- Example: Grass is wet $\Rightarrow$ It probably rained
- **Useful for diagnosis and planning**

Common KR Schemes

1. Logic-based Representations:

- Propositional Logic
- First-Order Logic (Predicate Logic)
- Higher-Order Logics

2. Structured Representations:

- Semantic Networks
- Frames and Scripts
- Conceptual Graphs

3. Rule-based Representations:

- Production Rules (IF-THEN)
- Decision Trees

4. Probabilistic Representations:

- Bayesian Networks
- Markov Models

Propositional Logic Overview

Basic Elements:

- **Propositions:** Statements that are true or false
- **Logical Connectives:** $\neg, \wedge, \vee, \Rightarrow, \Leftrightarrow$

Example Sentences:

- P : "It is raining"
- Q : "The ground is wet"
- $P \Rightarrow Q$: "If it is raining, then the ground is wet"

Advantages:

- Simple and well-understood
- Efficient inference algorithms exist

Limitations:

- Cannot represent objects and relations
- Limited expressiveness

First-Order Logic (FOL)

Extensions beyond Propositional Logic:

- **Objects:** Constants (John, Car1)
- **Relations:** Predicates (Owns, GreaterThan)
- **Functions:** father(John), age(x)
- **Quantifiers:** $\forall$ (for all), $\exists$ (there exists)

Example FOL Sentences:

- $\forall x \text{ Human}(x) \Rightarrow \text{Mortal}(x)$
- "All humans are mortal"
- $\exists x \text{ King}(x) \wedge \text{Lives}(x, \text{Castle})$
- "There exists a king who lives in a castle"

Advantage:

- Much more expressive than propositional logic
- Can represent complex relationships

Advantages of Knowledge-based Agents

Benefits:

- **Modularity:** Knowledge is separate from inference
- **Flexibility:** Easy to add new knowledge
- **Transparency:** Knowledge is explicit and inspectable
- **Reasoning:** Can explain decisions
- **General-purpose:** Same inference engine for different domains

Applications:

- Expert systems (medical diagnosis, financial advising)
- Planning and scheduling
- Natural language understanding
- Robotics and autonomous systems

Challenges and Limitations

Key Challenges:

① Knowledge Acquisition Bottleneck:

- Difficult to encode all necessary knowledge
- Requires domain experts

② Computational Complexity:

- Inference can be computationally expensive
- Exponential growth with KB size

③ Uncertainty and Incompleteness:

- Real world is uncertain
- Cannot encode everything

④ Brittleness:

- May fail on unexpected situations
- Lack of common sense reasoning

Solutions and Approaches:

- **Machine Learning Integration:**
 - Learn knowledge from data
 - Hybrid systems
- **Probabilistic Reasoning:**
 - Handle uncertainty explicitly
 - Bayesian approaches
- **Non-monotonic Reasoning:**
 - Retract conclusions when new info arrives
 - Default logic
- **Efficient Inference Algorithms:**
 - Optimized search strategies
 - Heuristic methods

Knowledge-based vs. Other Agents

Agent Type	Advantages	Disadvantages
Simple Reflex	Fast, simple	No memory, limited
Model-based	Handles partial observability	Limited reasoning
Goal-based	Flexible, can plan	Computationally expensive
Knowledge-based	Powerful reasoning, explainable	Complex, knowledge acquisition hard
Learning	Improves over time	Requires training data

Reference: Russell & Norvig (2020), Chapter 2

When to Use Knowledge-based Agents

Good Fit When:

- Domain knowledge is available and can be articulated
- Reasoning and explanation are important
- Rules and relationships are well-defined
- Transparency is required (e.g., medical, legal domains)

Poor Fit When:

- Knowledge is implicit or hard to articulate
- Real-time performance is critical
- Environment is highly uncertain
- Learning from data is more practical

Summary

Key Takeaways:

- Knowledge-based agents reason using explicit representations
- Architecture: Sensors $\rightarrow$ KB $\rightarrow$ Inference $\rightarrow$ Actuators
- Core operations: TELL and ASK
- Wumpus World demonstrates practical reasoning
- Multiple knowledge representation schemes available
- Challenges include knowledge acquisition and computational complexity

Next Steps

In upcoming lectures, we will explore:

- Propositional and First-Order Logic in detail
- Inference algorithms (Resolution, Forward/Backward chaining)
- Probabilistic reasoning methods

Emerging Directions:

① **Neuro-symbolic AI:**

- Combining neural networks with symbolic reasoning
- Best of both worlds

② **Knowledge Graphs:**

- Large-scale structured knowledge
- Google, Amazon, Facebook use extensively

③ **Explainable AI:**

- Increasing demand for transparent systems
- Knowledge-based approaches naturally explainable

④ **Common Sense Reasoning:**

- Building agents with human-like understanding
- Active research area

-  Russell, S., & Norvig, P. (2020). *Artificial Intelligence: A Modern Approach* (4th ed.). Pearson.
- Chapter 7: Logical Agents
 - Chapter 2: Intelligent Agents
-  Rich, E., Knight, K., & Nair, S. B. (2009). *Artificial Intelligence* (3rd ed.). McGraw-Hill.
- Chapter 5: Knowledge Representation
 - Chapter 6: Logic and Inference

Thank You!